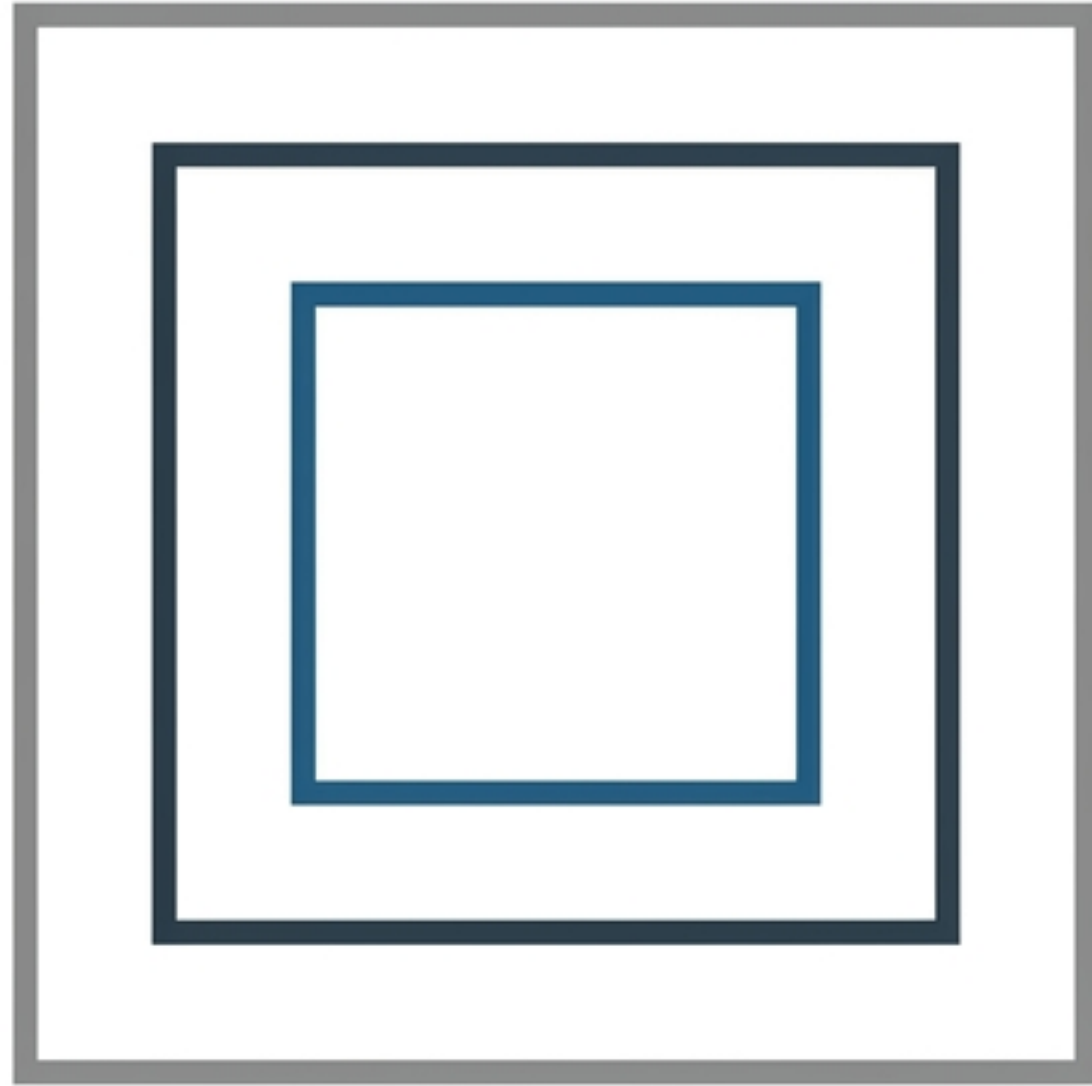




Module 1, Lesson 1

AI 101 for Mechanical Engineers

A Map of Related Ideas

Lesson Structure

1 System Taxonomies and Subsets

2 Generative Capabilities and Data Science

3 Applied Plant System Architectures

4 Terminology and Concept Verification

System Taxonomies and Subsets



Defining Artificial Intelligence Constraints

- Encompasses systems performing intelligent behavioral tasks.
- Includes learning-based operational approaches.
- Includes explicit rule-based automation approaches.
- Includes search and knowledge-based frameworks.

Artificial Intelligence

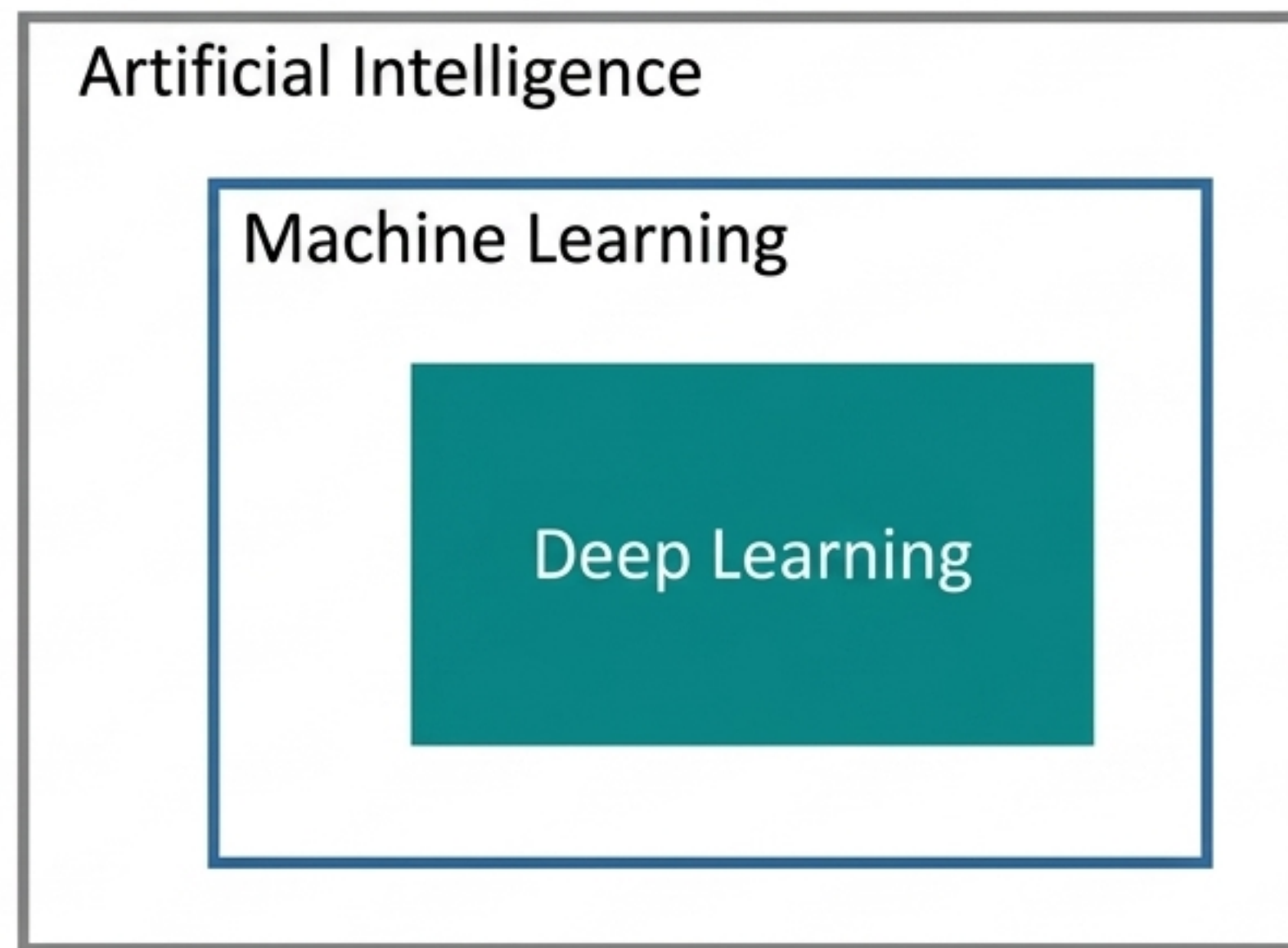
Rules

Search

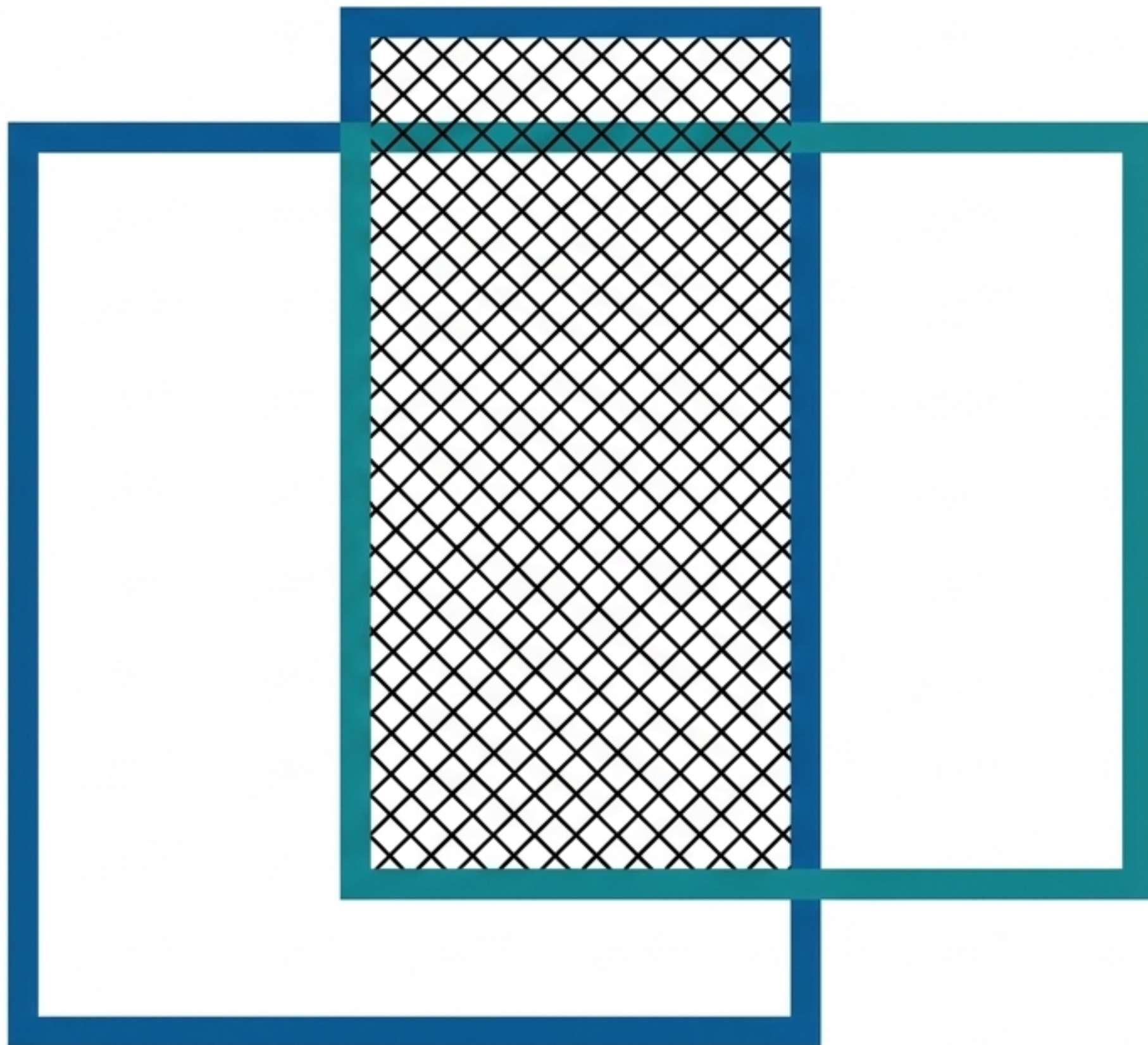
Learning
Methods

Parameters of Machine and Deep Learning

- Machine learning isolates patterns from data examples.
- Data determines specific system behavioral parameters.
- Deep learning relies on multi-layer neural networks.
- Deep learning functions as a strict subset.

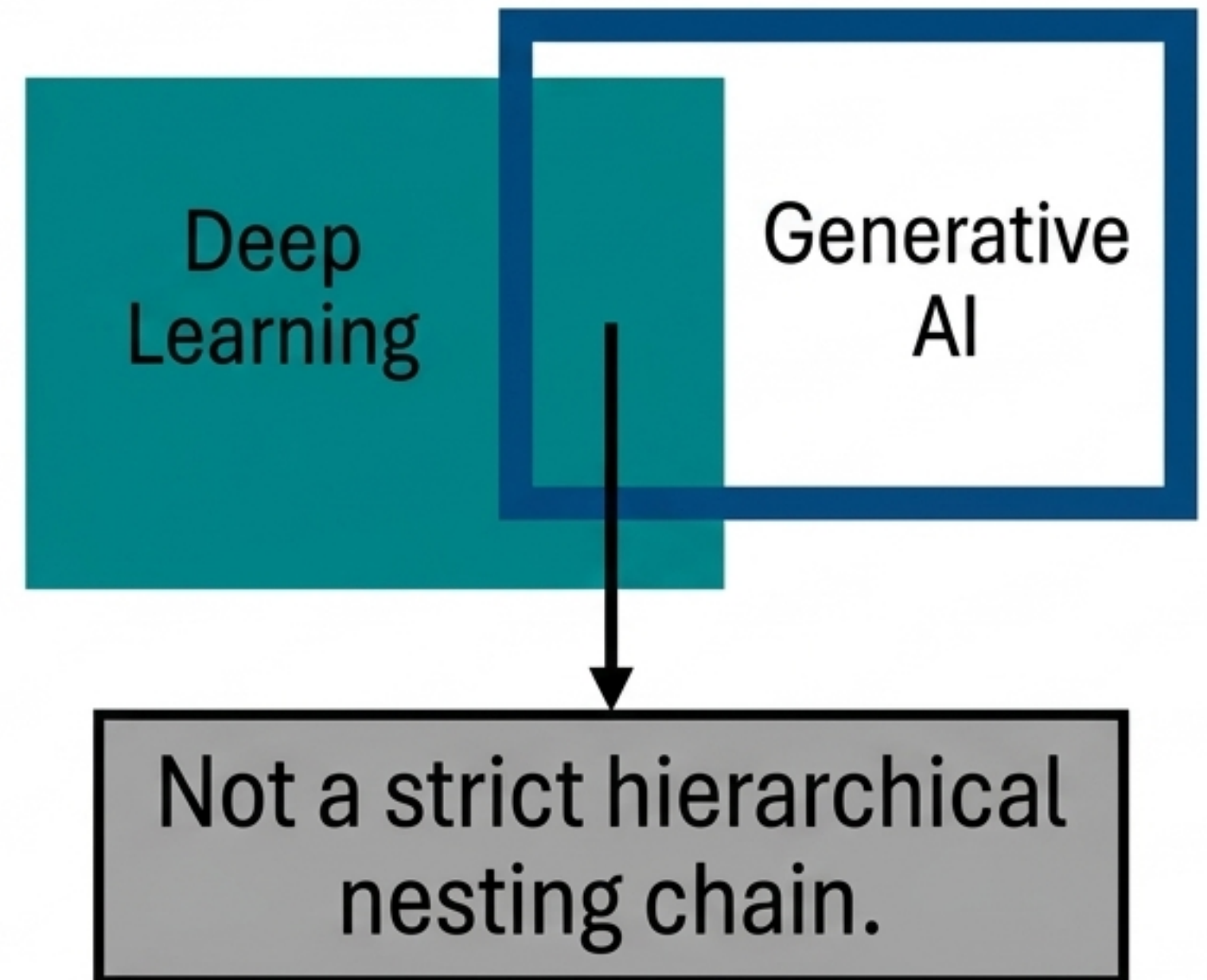


Generative Capabilities and Data Science



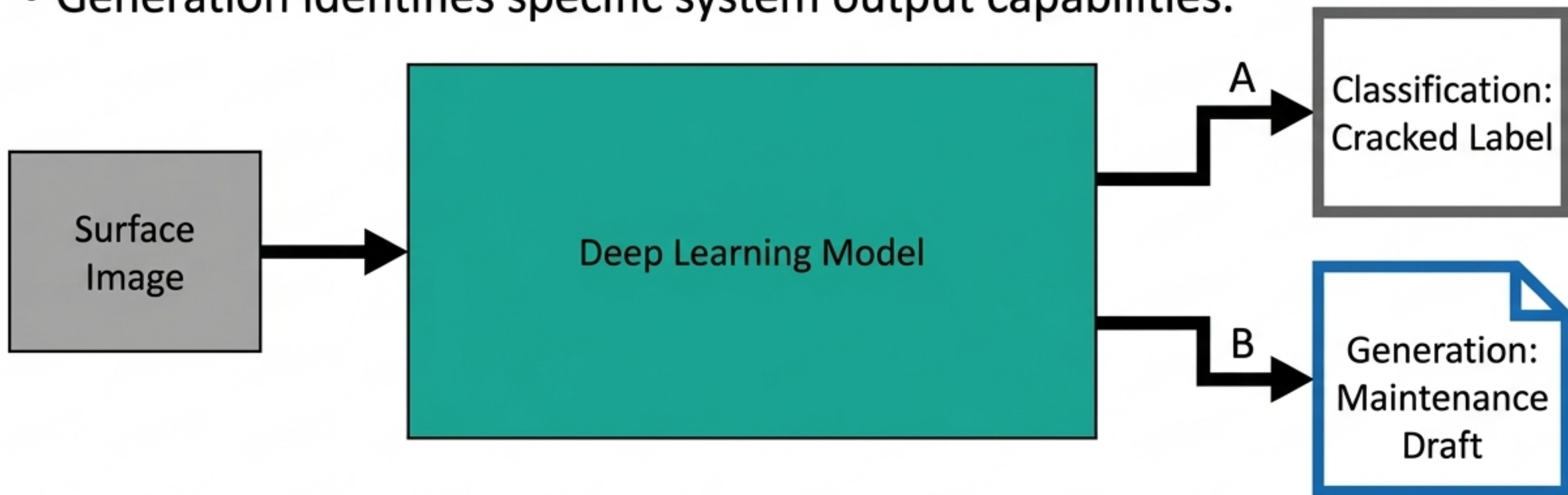
Generative System Capabilities

- Defines output capability, not a system hierarchy.
- Produces distinct target content (text, code, images).
- Functions independently of strict model classifications.
- Can operate utilizing deep learning frameworks.



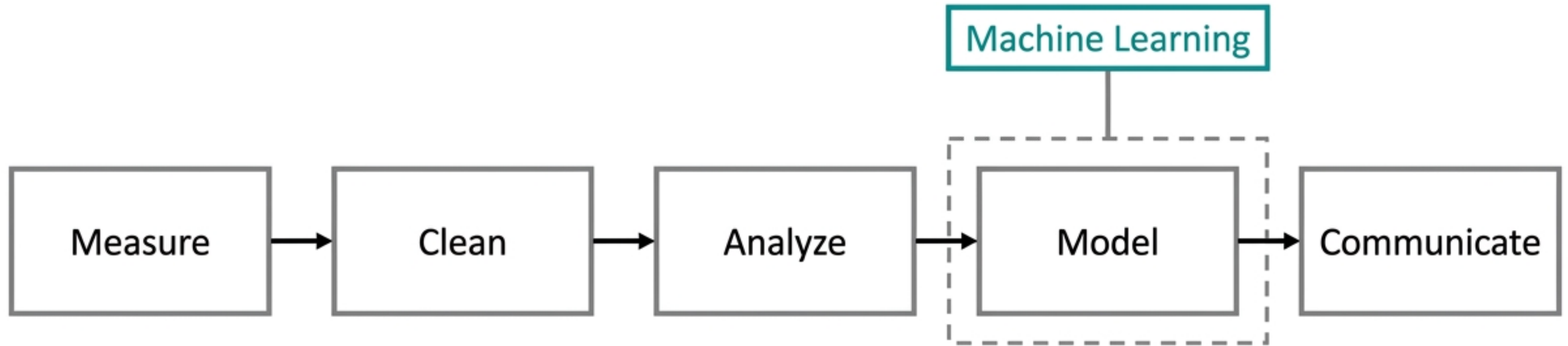
Distinguishing Classification from Generation

- Multi-layer networks execute non-generative classification tasks.
- Image classification requires no new content generation.
- Deep learning identifies the internal model family.
- Generation identifies specific system output capabilities.

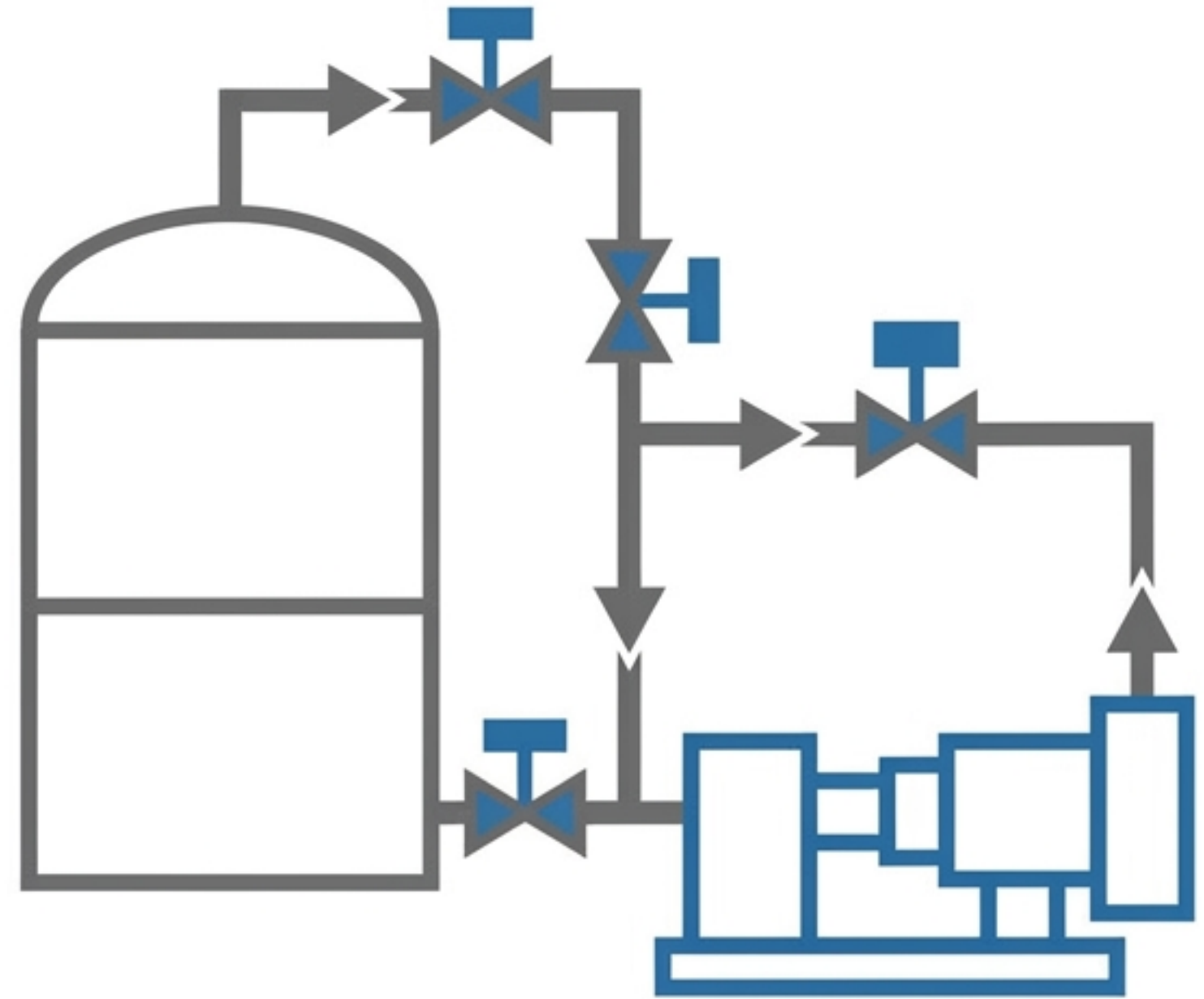


Scope of the Data Science Workflow

- Overlaps standard artificial intelligence frameworks.
- Requires structured mechanical data acquisition.
- Includes calibration and missing reading inspections.
- Processes modeling, analysis, and data communication.

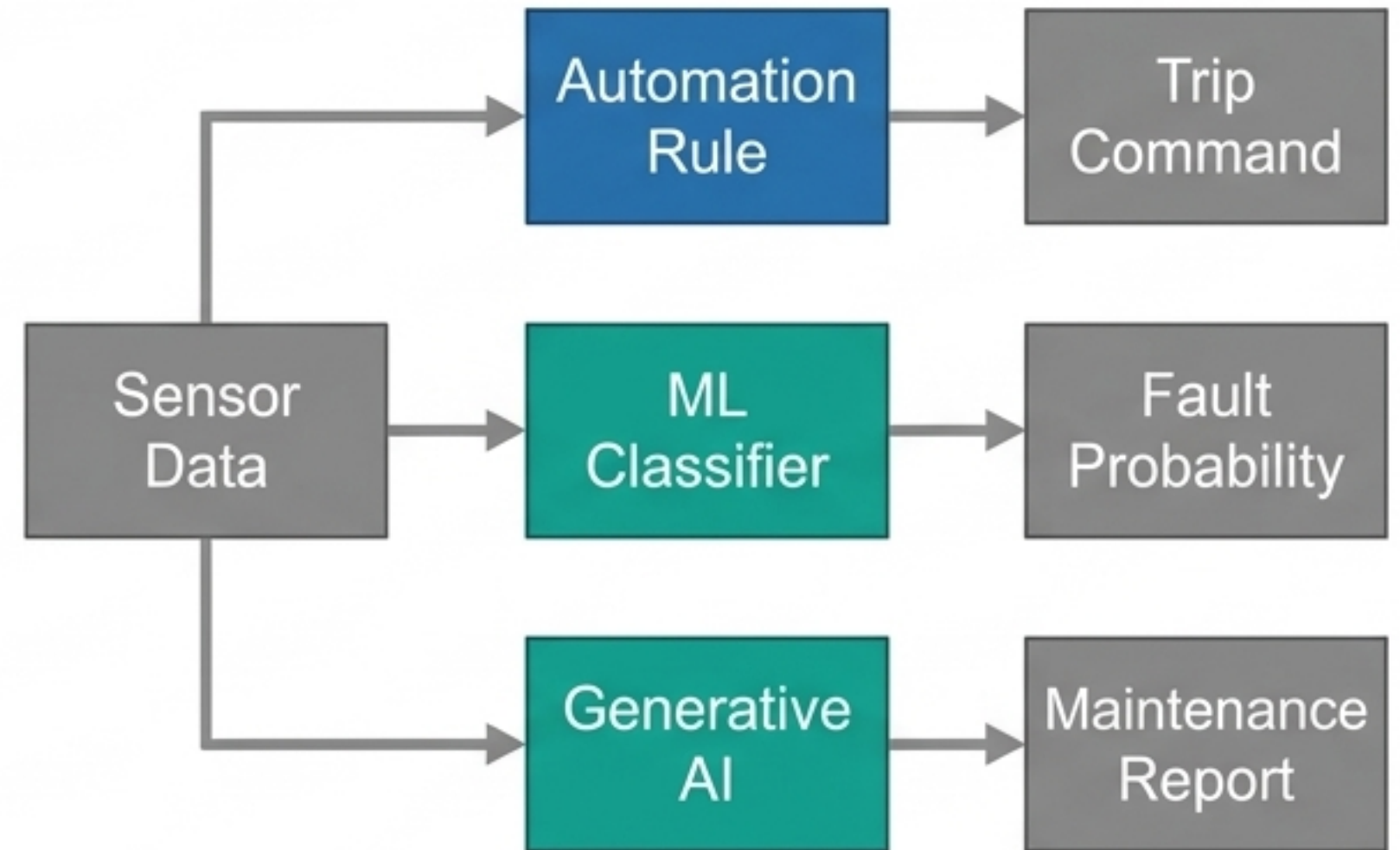


Applied Plant System Architectures



Parallel Architectures in Plant Environments

- Distinct intelligence systems operate simultaneously.
- Fixed over-temperature trips utilize automation rules.
- Learned classifiers estimate bearing-fault probabilities.
- Generative systems draft formal maintenance explanations.
- Each requires distinct operational evidence bases.

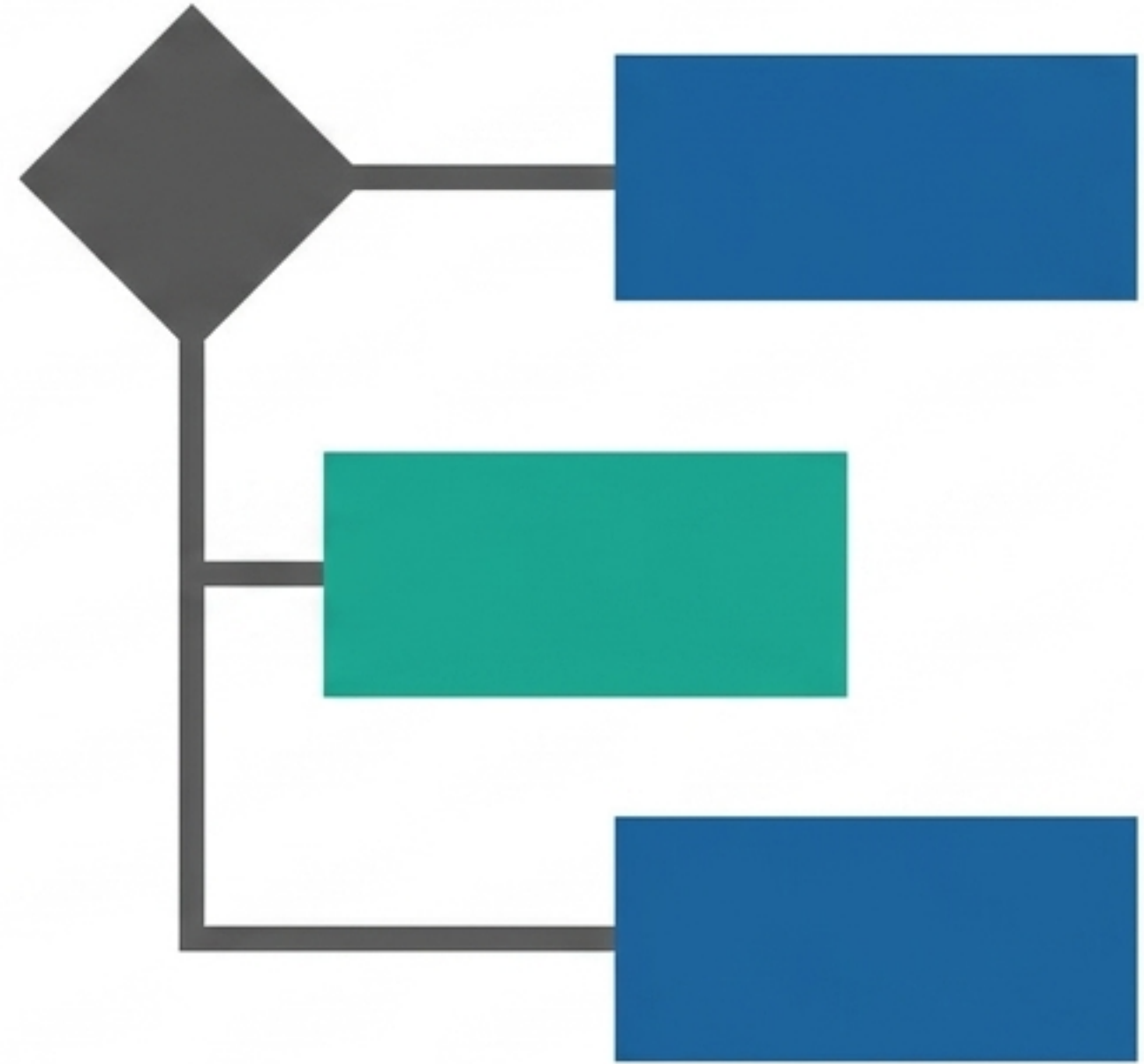


Consolidated Conceptual Taxonomy

- Table aligns core methodologies with industrial tasks.
- Specifies framework scope per defining terminology.
- Provides distinct mechanical engineering implementations.

Term	Scope Definition	Mechanical Engineering Example
Artificial Intelligence	Broad family of approaches	Planning or fault detection
Machine Learning	Learning from data examples	Fitted pump-power predictor
Deep Learning	Multi-layer neural network learning	Surface-defect classifier
Generative AI	Target content production	Draft formal maintenance text

Terminology and Concept Verification



Architectural Verification and Constraints

Verified Engineering Principles

- Machine learning constitutes one specific AI approach.
- Multi-layer networks routinely execute non-generative classification.

Validation Constraints

- Fluent outputs never guarantee engineering correctness.
- Synthetic evidence cannot validate physical mechanical equipment.

Key Points

- System Taxonomies: Machine and deep learning represent distinct nested subsets.
- Generative Capabilities: Generation defines output functionality, not the structural hierarchy.
- Applied Architectures: Rule-based, learning, and generative systems execute parallel responsibilities.
- Concept Verification: Fluency does not validate physical equipment or computational accuracy.